System and Devices (SYS 3)

Tutorial Lecture 3 : Static & Dynamic Scheduling

Dr Pengcheng Liu

True/False

Which of the following statements is/are FALSE?

1. In a dynamic scheduled processor, every execution unit writes the result and the name of the reservation station waiting for the result on to the CDB.
2. In a speculative dynamic scheduled processor, we can issue an instruction if there is an empty reservation station even though operands are not available.
3. If the register status indicator value of the register is 0, then it means the operand is available in Register File.
4. In a dynamic scheduled pipeline, instructions are issued in order, executed out of order, completed out of order and committed in order.

(A) 2 and 3 only (B) 1 and 2 only (C) 1 and 4 only (D) 3 and 4 only

Tomasulo's Algorithm

Consider an instruction pipeline of an issue width of 1 that uses Tomasulo's algorithm with two reservation station per functional unit. There is one Integer Mul Unit, one Integer Div unit and one Integer Add unit all connected to a single CDB. There is an arbitration mechanism for resolving CDB entry collision. Preference is given in round robin fashion. The functional units are not pipelined. An instruction waiting for data on CDB can move to its EX stage in the cycle after the CDB broadcast.

Assume the following information about functional units.

Functional unit type	**Cycles in Execution stage**
Integer Mul	*4*
Integer Div	*8*
Integer Add	*1*

Complete the following table using Tomasulo's algorithm with the above specifications. Fill in the cycle numbers in each pipeline stage for each instruction, and indicate where its source operands are read from (use RF for register file, ROB for reorder buffer and CDB for common data bus).

Tomasulo's Algorithm

#	Instruction	Issue	Src_Op1	Src_ Op2	EX	WB	Commit
1	DIVI R4, R4, #12	1	RF	Imm	2	10	11
2	MUL R2, R6, R12	2					
3	DIV R1, R1, R2	3					
4	ADD R5, R1, R3						
5	ADDI R7, R2, #4						
6	ADD R5, R6, R7						
7	ADDI R8,R8, #24						
8	ADD R9, R6, R8						
9	MUL R5, R5, R10						
10	ADD R6, R8, R5						